

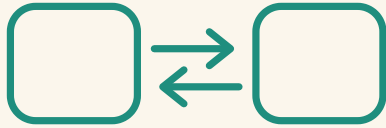
Foundations Flashcards

CCA·F — the 10 building blocks · for dyslexic beginners





API



IN PLAIN WORDS

Two programs talking — you send a request, you get a response.

PICTURE IT

Ordering at a counter: you ask, the kitchen hands it back.

EXAMPLE

Your code asks the Claude API "summarize this" → it returns the summary.

WHY IT MATTERS

Every Claude app is just your code talking to the API.

⚠ WATCH OUT

The API key is secret — never put it in your code.



JSON



IN PLAIN WORDS

A simple text way to write structured data.

PICTURE IT

A labelled form — every field has a name and a value.

EXAMPLE

`name = "Dan", score = 20` — wrapped in curly braces `{ }`.

WHY IT MATTERS

It's how data moves in and out of the API.

⚠ WATCH OUT

One missing comma or quote and it won't parse.



Schema



IN PLAIN WORDS

The rules for the shape of data — which fields, what type.

PICTURE IT

A cookie cutter — the data has to fit the shape.

EXAMPLE

An invoice schema requires total (number), allows notes (text).

WHY IT MATTERS

Makes Claude's output predictable and catches bad data.

⚠ WATCH OUT

Too strict breaks on messy data; too loose lets junk through.



Context window



IN PLAIN WORDS

The model's short-term memory — how much it holds at once.

PICTURE IT

A desk that only fits so many papers — extras fall off.

EXAMPLE

Paste a huge doc + long chat → the earliest text drops out.

WHY IT MATTERS

If info falls out, the model simply forgets it.

⚠ WATCH OUT

Overstuffing buries key facts (lost-in-the-middle).



Tool



IN PLAIN WORDS

An action you let the model call to do or fetch something.

PICTURE IT

Giving someone a phone and a calculator.

EXAMPLE

A `lookup_order` tool lets Claude check a real order, not guess.

WHY IT MATTERS

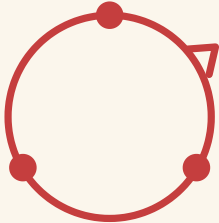
Tools turn a chat model into an agent that can act.

⚠ WATCH OUT

A vague description makes Claude pick the wrong tool.



Agent



IN PLAIN WORDS

Model + tools + a LOOP: think → use tool → read result → repeat.

PICTURE IT

A worker who keeps going until the job is done.

EXAMPLE

A support agent checks an order, decides a refund, escalates.

WHY IT MATTERS

Agentic Architecture is the biggest exam domain.

⚠ WATCH OUT

No clear stop rules and the loop can run away.



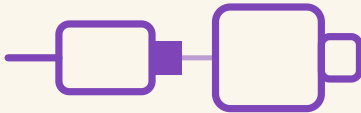
MCP

IN PLAIN WORDS

A standard plug connecting Claude to outside tools and data.

PICTURE IT

A USB port — anything that fits the standard plugs in.



EXAMPLE

An MCP server exposes a doc catalog and a search_docs tool.

WHY IT MATTERS

Safe, reusable access to real systems.

⚠ WATCH OUT

Don't mix tools (actions), resources (read-only), prompts (workflows).



Prompt



IN PLAIN WORDS

The instruction you give the model.

PICTURE IT

A recipe — clear steps give a good dish; vague gives a mess.

EXAMPLE

"Extract the invoice total as JSON. If missing, return null."

WHY IT MATTERS

Prompts shape behaviour, output format and error handling.

⚠ WATCH OUT

Prompt-only rules aren't enough for safety-critical — use code.



Resource



IN PLAIN WORDS

Read-only context exposed by MCP — a doc, catalog or list.

PICTURE IT

A reference shelf — you read it, you don't change it.

EXAMPLE

A refund-policy resource lets Claude read the current policy.

WHY IT MATTERS

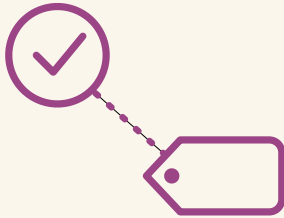
Trusted context without messy searching.

⚠ WATCH OUT

Don't use a tool (an action) when a read-only resource fits.



Provenance



IN PLAIN WORDS

Tracking where each fact or claim came from.

PICTURE IT

A receipt — proof of where each thing came from.

EXAMPLE

"Refund policy allows 30 days" → source: `refund-policy.md`.

WHY IT MATTERS

Production systems need source-tracking so humans can audit.

⚠ WATCH OUT

An answer with no source can't be verified.